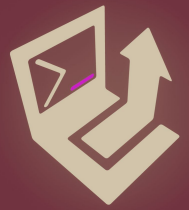
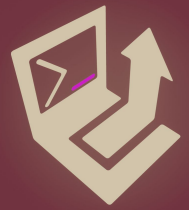
DevLUp FSU GBM #7





Materials in Blender

29th October, 2026

Meeting Schedule

Date	GBM	GBM Title	Secondary Event	Presenter
		(No Meeting)	ACM Sep 4th	
17 Sep	1	First Meeting		Everyone
24 Sep	2	General introduction to Unity		Yusong
1 Oct	3	Game play and level design		Mitchell
8 Oct	4	Roblox development (free development)		Ashi
15 Oct	5	Modeling in Blender	Game Jam?	Sean
22 Oct	6	Pixel art		Ashi
29 Oct	7	Materials in Blender		Kaydence
5 Nov	8	voice acting in games		Laura
12 Nov	9	Music in games		Marco
19 Nov	10	The math behind game dev		Miles
3 Dec	11	End of Fall celebration		Everyone

Game Jam Reminder!!!!!




#👀showoff

Lesson Plan

- ➔ Making Game-Ready Models
- ➔ Without UV Unwrapping
- ➔ Triplanar Mapping
- ➔ Game-Ready Materials
- ➔ UV-Unwrapping
- ➔ Blender Texture Editor
- ➔ Material Maps
- ➔ Where to Get Materials



Download Berry

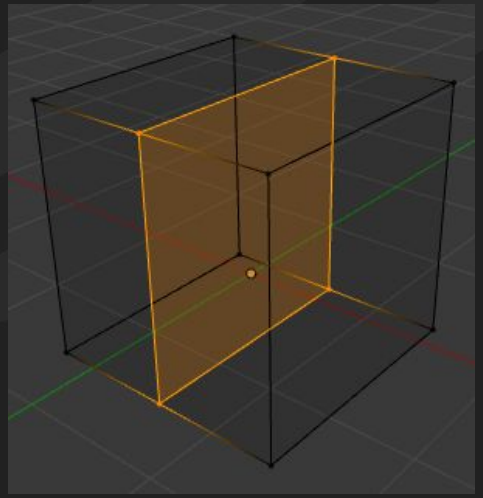
https://swipiverse.neocities.org/berry

What Are Materials?

- Color information
- Texture information
- Shading information

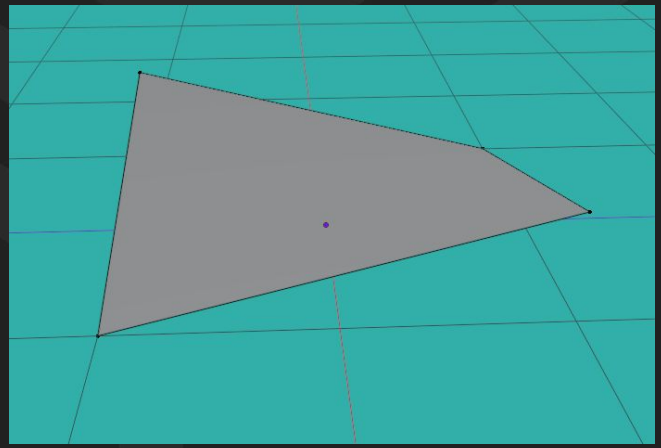
Making Game-Ready Models

- Good topology = Better materials
- Ensure no duplicate faces:
  - Shift+A -> M -> “Merge by Distance”
- Check for non-manifold/hidden faces
  - Non-manifold: can not exist in real life
  - Internal faces cause weirdness with UVs



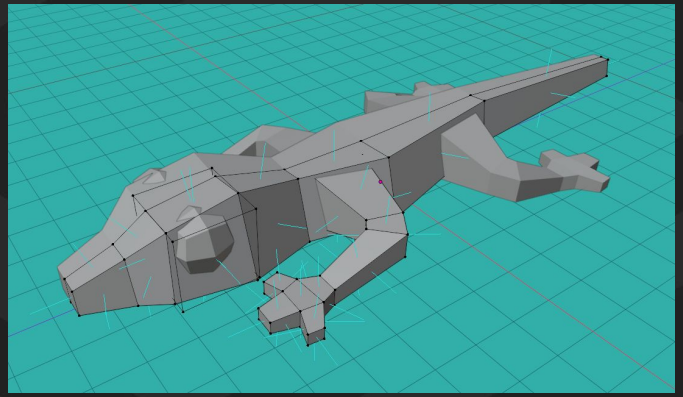
Making Game-Ready Models

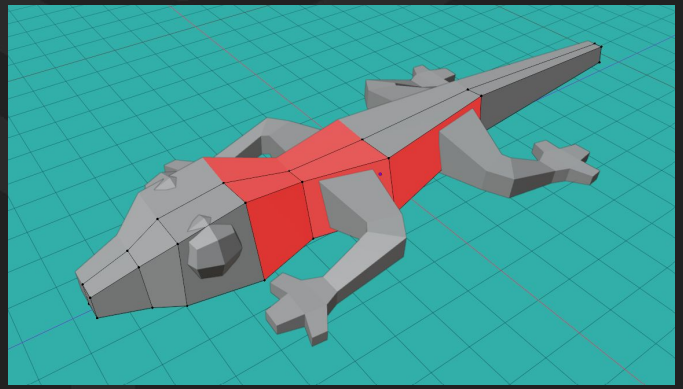
- Non planar faces: not flat
  - Not the end of the world by try to avoid. Make weird normals
  - Avoid n-gons
- If intended to deform: watch for pinch/stretch points



Normals

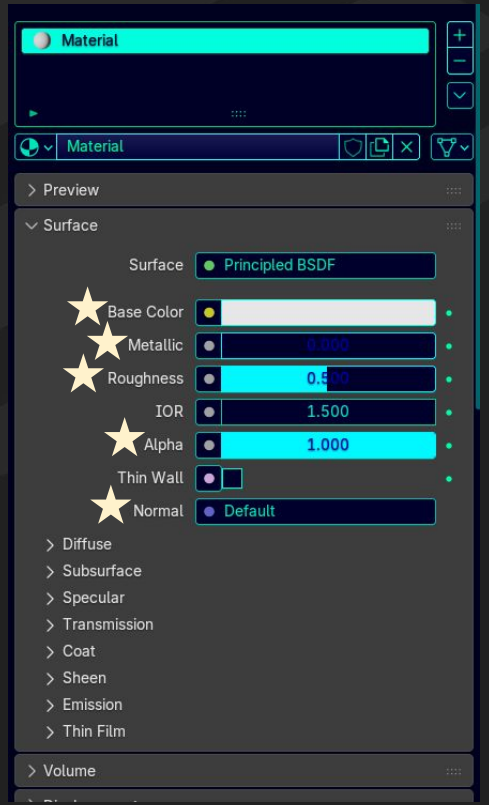
- Determine how model is shaded
  - “Shade Smooth” and “Shade Flat”
- What is considered “front” or “back” of face
  - Normally back of face is not rendered
- RECALCULATE YOUR NORMALS
  - Ctrl+A -> Shift+N





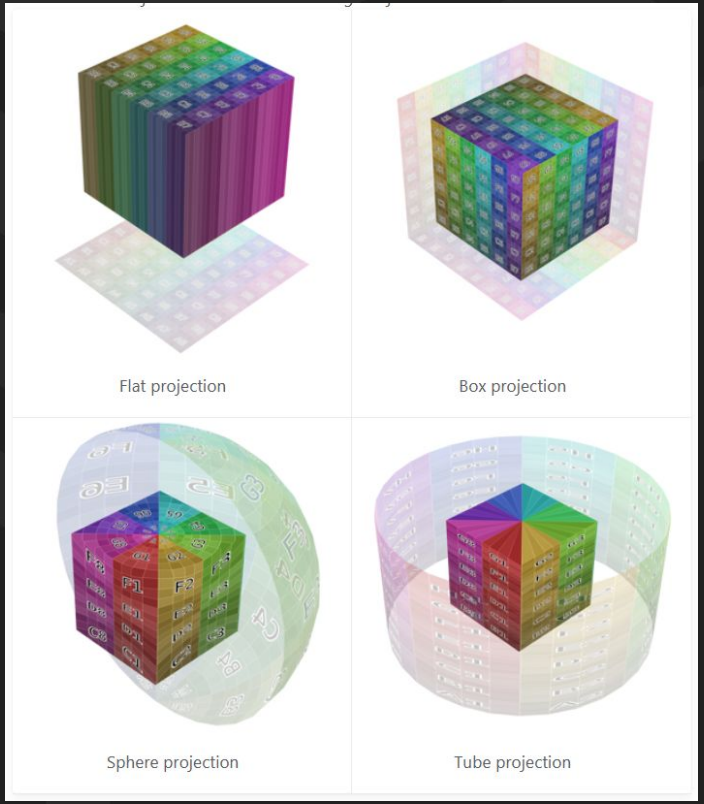
Material Menu in Blender

- Many different types of materials
- Stick with basic BSDF (godot/unity)
- Shaders don’t translate
- Base color = albedo
  - Single color or texture map
- Good practices
  - More materials = more draw calls (more work)
  - Smaller prop = smaller texture



Non-UV Unwrapping

- Ways of projecting image textures
  - Doesn’t translate directly but can be baked into uvs
  - [image texture on blender docs]

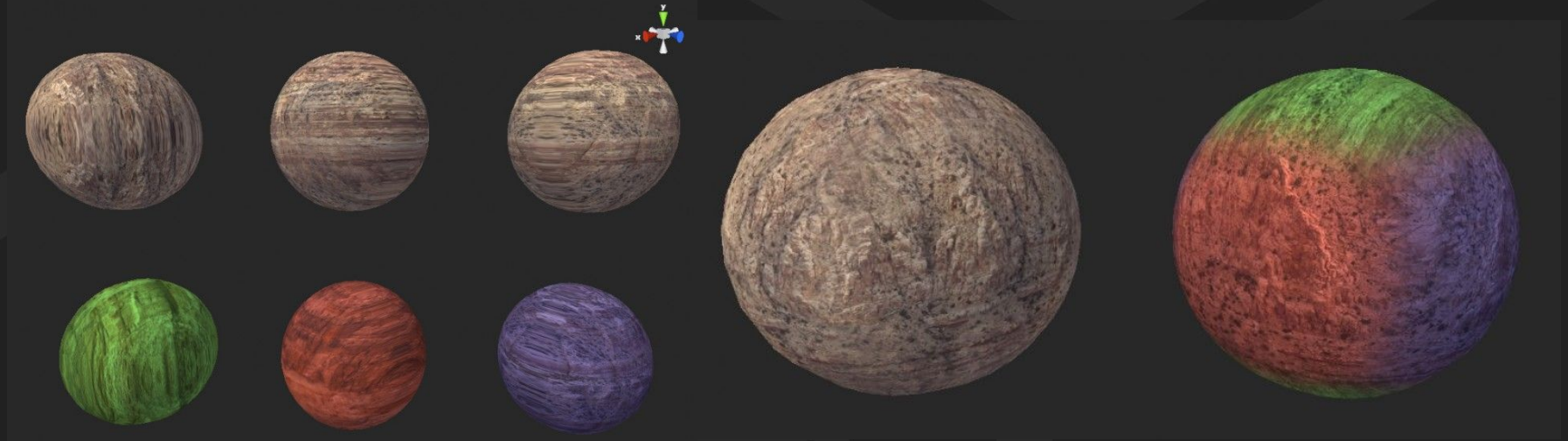


Practice on Berry

Put your Berry in Showoff

Triplanar Mapping

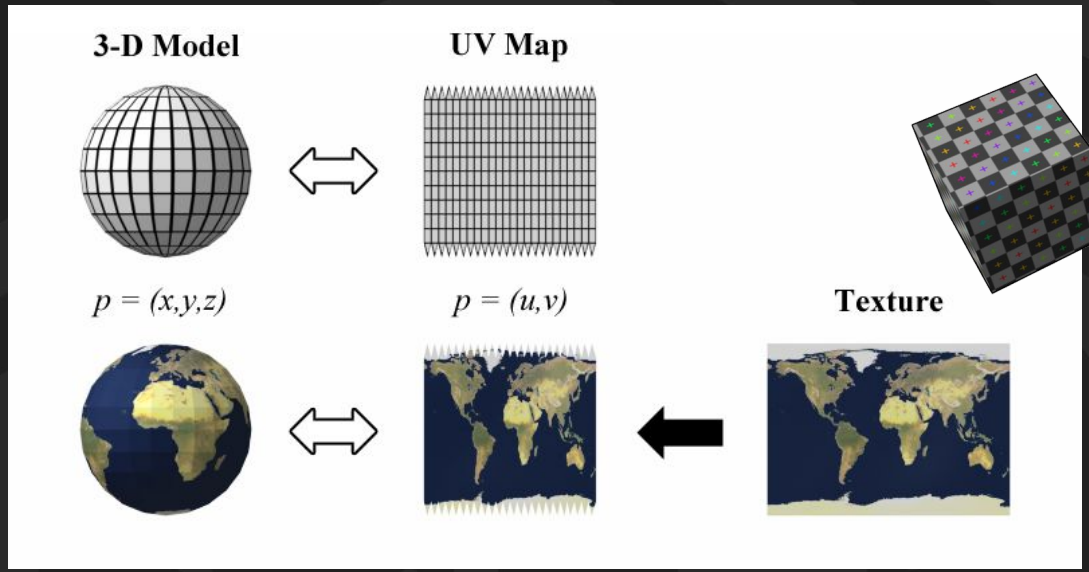
- Flat projection on front, sides, top
- In-engine way of avoiding UVs
- Unity implementation by hand

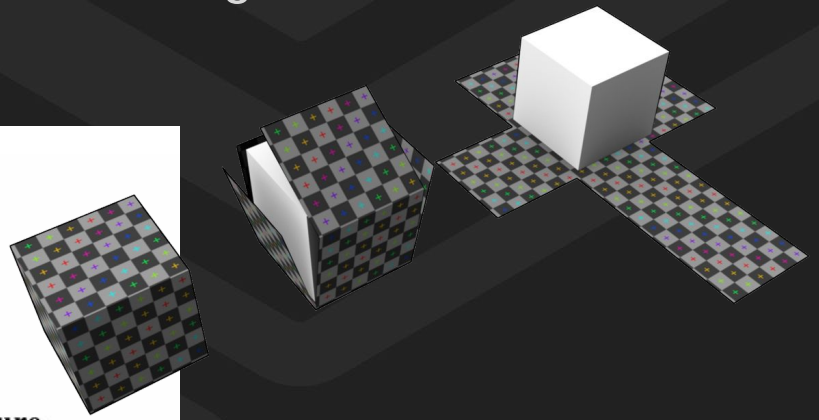


The Almighty UV

What are UVs?

- How do you represent a 3d model as a 2d image?



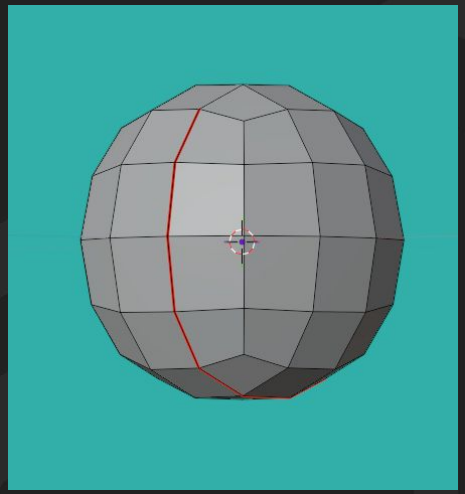


What are UVs?

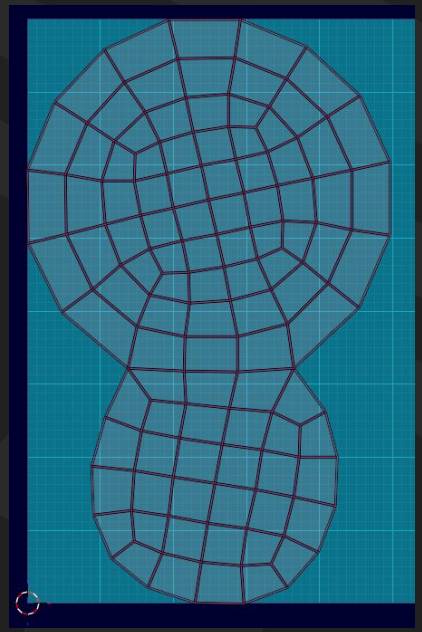
- UVs are stored with mesh
- Can be moved around on a texture
- Can technically be edited with code



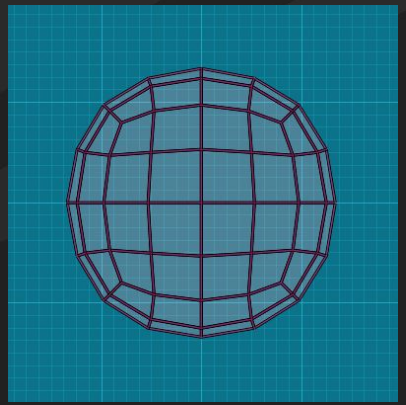
Sphere with seam



Seams + unwrap
minimum stretch

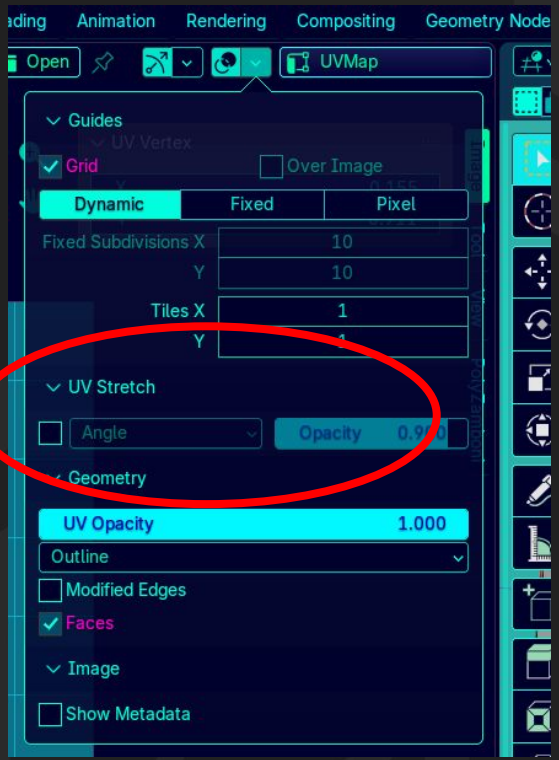


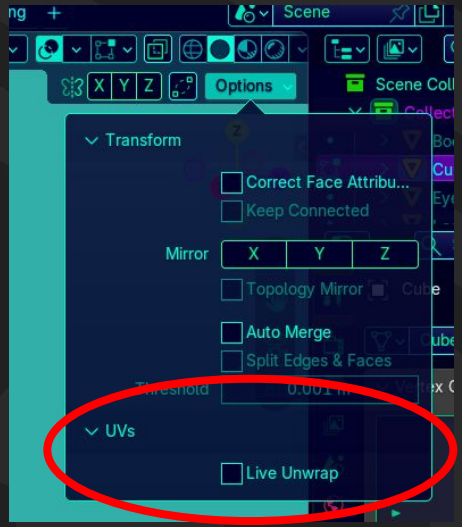
Project from view



UV Unwrapping Demonstration

Some Useful UV Things





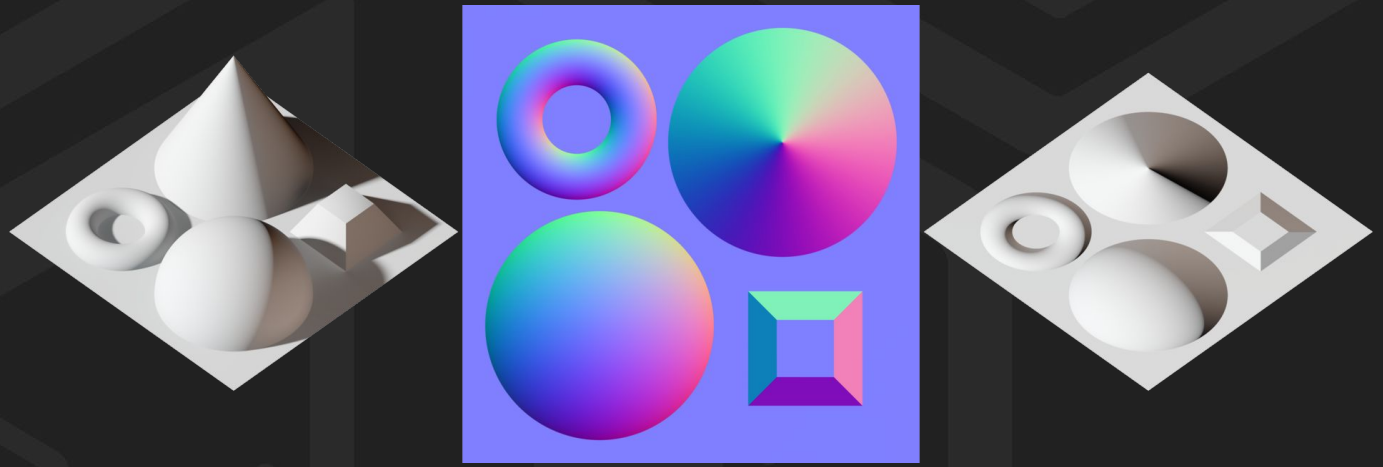


Texture Paint Demonstration

Texture Images as Information

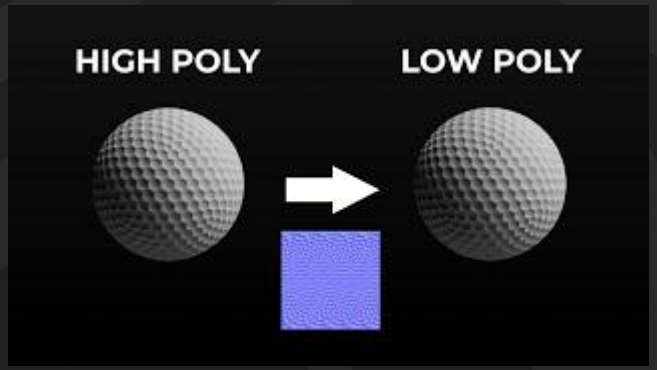
Normal Maps

- Lighting information without the topology
- Lowers poly count but falls apart upclose
- Best for small details



Baking Normal Maps

- High poly meshes are baked into normal maps
- Maps applied to low poly meshes





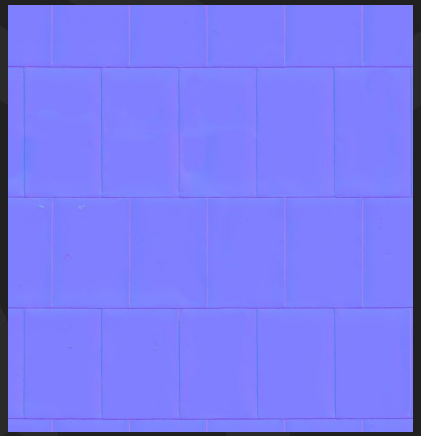
Other Maps

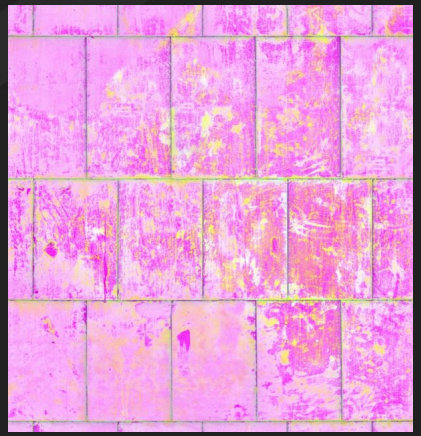
- https://polyhaven.com/a/metal_plate_02
- Roughness map, metallic map, alpha map





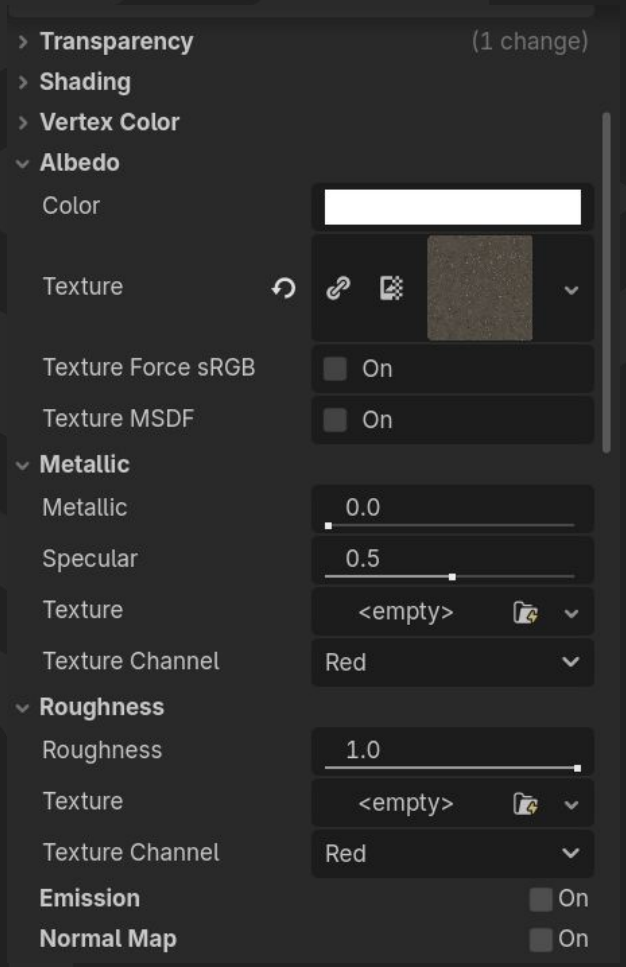






Other Maps

- Supported by game engines



Where to Find/Make Textures

PBRs

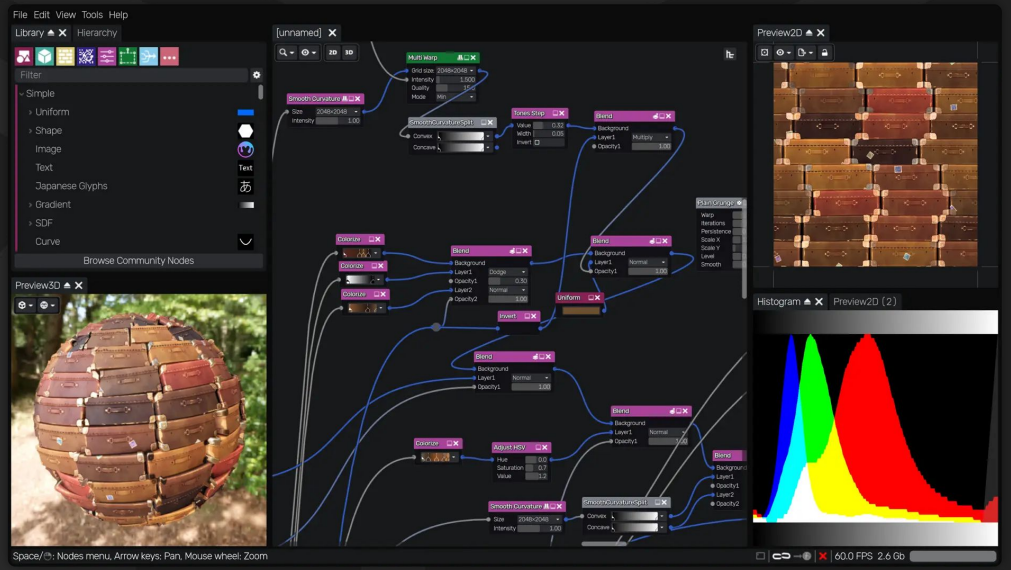
- Physically-based rendering
- https://polyhaven.com/textures

They literally just scan things from irl and make high quality textures from them



Material Maker

- Procedurally generate PBRs
- https://www.materialmaker.org/



Final Berry Showoff

Thanks for hanging out

